

Lecture 2: Normal Linear Models — Structure, Types and Matrix Form

MATH3701 Statistical Modelling

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Course notes: MATH3701 Statistical Modelling notes, Chapter 2

Note: This is revision material. The matrix formulation will be needed for GLM theory in later lectures.

The Normal Linear Model

For n observations, the **general linear model** is:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

- $\mathbf{Y} = (Y_1, \dots, Y_n)^T$: response vector
- $\mathbf{X}$: $n \times p$ **design matrix** (known)
- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$: parameter vector (unknown)
- $\boldsymbol{\epsilon}$: independent errors with constant variance σ^2

Equivalently:

$$Y_i \mid \mathbf{x}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{x}_i^T \boldsymbol{\beta}, \sigma^2)$$

Types of Normal Linear Models

Parameters	Explanatory variables	Model name
$p = 2$	1 quantitative	Simple linear regression
$p > 2$	> 1 quantitative	Multiple linear regression
$p = 2$	1 binary	Two-sample t -test
$p = k$	1 qualitative (k levels)	One-way ANOVA
$p > k$	Qualitative only	Multi-way ANOVA
$p > 2$	Mixed	ANCOVA

Key insight: These are all *special cases* of $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.

The Design Matrix

Recipe for building $\mathbf{X}$:

- ① Column of 1s (intercept)
- ② One column per quantitative variable
- ③ For a qualitative variable with k levels: $k - 1$ dummy indicator columns (one level = reference)
- ④ Interaction columns = element-wise products

Example (simple regression with $n = 4$):

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ 1 & x_4 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$\mathbf{X}$ must have full column rank p for a unique solution to exist.

Least Squares Estimation

Minimise the residual sum of squares (RSS):

$$\text{RSS}(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

Setting $\partial \text{RSS} / \partial \boldsymbol{\beta} = 0$ gives the **normal equations**:

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}$$

Least squares (OLS) estimator:

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}}$$

Properties: $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ (unbiased), $\text{Var}[\hat{\boldsymbol{\beta}}] = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$.

Hat Matrix, Fitted Values, Residuals

Fitted values:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \underbrace{\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T}_{\mathbf{H}} \mathbf{Y}$$

H is the **hat matrix**: symmetric, idempotent ($\mathbf{H}^2 = \mathbf{H}$), $\text{tr}(\mathbf{H}) = p$.

Residuals: $\mathbf{r} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$

Error variance estimate:

$$\hat{\sigma}^2 = \frac{\mathbf{r}^T \mathbf{r}}{n - p} \quad (\text{unbiased: } \mathbb{E}[\hat{\sigma}^2] = \sigma^2)$$

The hat matrix will reappear in Chapter 13 as the *smoothing matrix*.

Fitting in R

```
# Fit the model
fit <- lm(y ~ x1 + x2, data = mydata)

# Key output
summary(fit)      # coefficients, SE, t-tests, R^2
anova(fit)        # analysis of variance table

# Fitted values and residuals
fitted(fit)
residuals(fit)

# Compare nested models
fit0 <- lm(y ~ 1, data = mydata)
anova(fit0, fit)   # F-test

# Access the design matrix
model.matrix(fit)
```

Nested Model Comparison: The F-Test

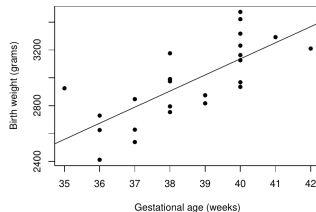
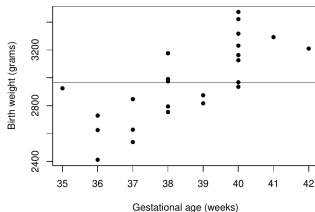
For nested models $M_0 \subset M_1$ with $r_0 > r_1$ residual degrees of freedom:

$$F_{01} = \frac{(\text{RSS}_0 - \text{RSS}_1)/(r_0 - r_1)}{\text{RSS}_1/r_1} \sim F_{r_0-r_1, r_1} \quad \text{under } H_0$$

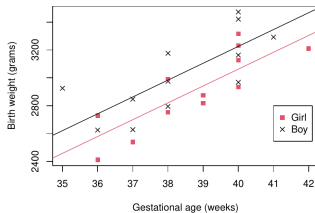
- Large $F \Rightarrow$ small p -value $\Rightarrow$ prefer M_1
- Small $F \Rightarrow$ large p -value $\Rightarrow$ use simpler M_0

Note: The F -test has a direct analogue in GLMs via the deviance ratio (Chapter 4).

Example: Nested Models for Birthweight Data

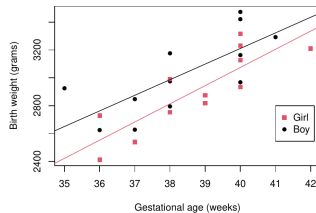


Model 0: intercept only



Model 2: + sex

Model 1: + age



Model 3: + age $\times$ sex

F -tests decide which additional terms are justified by the data.

Check the model assumptions by plotting residuals $r_i = y_i - \hat{y}_i$:

- ➊ **Residuals vs fitted:** Should show no pattern (tests linearity and constant variance)
- ➋ **Normal Q-Q plot:** Residuals should lie on a diagonal line (tests normality)
- ➌ **Scale-location:** Should be flat (tests homoscedasticity)
- ➍ **Residuals vs leverage:** Identifies influential observations

In R: `plot(fit)` produces these four plots automatically.

Key points:

- The normal linear model: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- The design matrix $\mathbf{X}$ encodes all model structure
- OLS estimator: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$
- The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ maps data to fitted values
- Nested models compared via F -test
- Residual plots check model assumptions

Next lecture: More on normal linear models